

ROOTPRINT

The Trust Layer for Kashmiri Craft

A voice-first AI system that turns oral craft knowledge into a scannable, verifiable, buyer-ready provenance identity.

Hackathon Product Blueprint | MVP Scope | Winning Pitch Narrative

1. The Big Idea

One sentence

Rootprint gives every Kashmiri craft product a living digital passport: the artisan speaks, AI structures the knowledge, the system creates a tamper-aware QR identity, and the buyer can scan, verify, listen, ask questions, and trust the product instantly.

Rootprint is not another marketplace. It is not another tourism listing. It is not a generic AI chatbot.

Rootprint is the proof infrastructure for Kashmiri craft.

Today, a fake “Kashmiri pashmina” can use the same words as a real one: handmade, authentic, heritage, pure, original. The buyer sees a label. The artisan has a lifetime of knowledge. There is no bridge between the two.

Rootprint builds that bridge.

The product begins with one simple action: an artisan, shop owner, cooperative, or craft student records a spoken story about a product or craft process. Rootprint converts that conversation into a structured **Craft Passport** containing origin, technique, material, lineage, media evidence, and a public QR page. The QR code travels with the product as its digital identity.

The winning frame

If Kashmir is globally known for craft, Kashmir should also own the digital standard for proving craft authenticity. Rootprint is that standard.

2. Why This Hits Hard for a Kashmir Jury

A jury at NIT Srinagar does not need to be convinced that Kashmiri craft matters. They need to be convinced that the team has found a technical wedge that is local, urgent, scalable, and demoable.

What the jury cares about	How Rootprint answers it
Local relevance	Starts with pashmina, carpets, papier-mache, walnut wood, copperware, embroidery, and craft clusters that Kashmir already owns in global memory.
Real impact	Gives artisans a way to preserve knowledge, prove authenticity, and increase trust with buyers.
Technical quality	Uses speech processing, LLM extraction, structured data, evidence scoring, QR identity, public verification pages, and an interactive buyer AI.
Feasibility	A focused MVP can be built in a hackathon without pretending to solve all certification law on day one.
Scalability	Once the passport layer exists, commerce, tourism, export documents, workshops, and APIs become modules on top.

Line to make the room pause

“If we do not build the source of truth for Kashmiri craft from Kashmir, the world will keep selling Kashmir without Kashmiri makers at the center.”

3. The Problem

3.1. The Trust Gap

The craft market has a brutal information problem. The people with the deepest knowledge have the least digital proof. The people selling imitations often have better packaging, better English, better SEO, and better access to buyers.

1. **Buyers cannot verify authenticity.** A tourist, online buyer, retailer, or collector cannot easily know whether a product is handmade, where it came from, or who made it.
2. **Artisans cannot carry their story with the product.** Once a shawl, carpet, box, or carved object leaves the workshop, the maker’s identity disappears.
3. **Knowledge is trapped in memory.** Techniques, motifs, family histories, tools, material choices, and process details are rarely captured in structured form.
4. **Middlemen benefit from opacity.** When proof is missing, margins flow to whoever controls the buyer relationship, not necessarily the maker.
5. **Fake products dilute the Kashmir brand.** Every counterfeit product weakens the premium that genuine Kashmiri artisans should command.

Kashmiri craft does not only need more buyers. It needs memory, identity, and proof that can travel with the product.

4. The Product: Rootprint Craft Passport

4.1. What the User Experiences

1. **The artisan speaks.** They record a short voice note or interview in Kashmiri, Urdu, Hindi, or English.
2. **Rootprint understands.** AI transcribes, translates, and extracts craft-specific facts into a structured record.
3. **The record is reviewed.** The artisan/team can correct names, places, techniques, materials, and evidence.
4. **The passport is minted.** Rootprint generates a unique Passport ID, QR code, score, and public page.
5. **The buyer scans.** The product page opens instantly, showing the maker, process, evidence, and story.
6. **The buyer asks.** A mini AI guide answers questions from the verified record: “Why is this expensive?”, “What is kani weaving?”, “How do I know this is handmade?”

4.2. What a Craft Passport Contains

- | | |
|---------------------------------------|----------------------------------|
| • Artisan or business identity | • Family/apprenticeship lineage |
| • Craft category | • Process summary |
| • Village, district, or craft cluster | • Audio transcript excerpt |
| • Product details | • Photos or short video evidence |
| • Technique names | • Passport completeness score |
| • Tools used | • Evidence level |
| • Material provenance | • QR code and public URL |

The product metaphor

For buyers, Rootprint feels like scanning a museum label, certificate, and maker story at the same time. For artisans, it feels like speaking once and receiving a professional digital identity.

5. The “Head Spin” Features

The MVP should not be a basic CRUD app with QR codes. It needs two or three moments that make judges sit forward.

5.1. 1. Voice-to-Passport in Front of the Jury

The team records a 60–90 second artisan-style answer live:

“My name is ...I am from ...This is a handwoven pashmina shawl. We use ...The technique came from ...The material is sourced from ...”

Rootprint turns it into a polished Craft Passport with structured fields, story text, score, and QR code.

Judge reaction target: “They created an authenticity page from speech in under two minutes.”

5.2. 2. Ask the Craft: Buyer AI on the QR Page

After scanning the QR page, the judge can ask questions:

- “Why is this product valuable?”
- “What technique was used?”
- “What evidence supports the claim?”
- “Explain this to a tourist in simple words.”
- “Generate a luxury product description for an international buyer.”

The AI answers only from the passport record, making the product page interactive and trustworthy.

Why this is powerful

Most QR certificates are dead pages. Rootprint’s passport is conversational. The product can explain itself.

5.3. 3. Fake-vs-Rootprint Scan Moment

Show two product cards:

1. A generic tag saying “Authentic Kashmiri Handmade” with no Rootprint QR.
2. A Rootprint tag with a QR code and Passport ID.

When the first is scanned or searched, the app shows: **No verifiable Rootprint found. Ask the seller for proof.**

When the second is scanned, it opens the full passport.

Judge reaction target: “This could change consumer behavior. People will start asking for Rootprint.”

5.4. 4. Evidence Score, Not Blind Certification

The system does not make reckless legal claims. It shows a transparent score:

- Identity provided
- Location provided
- Technique specificity
- Material detail
- Media evidence
- Human review
- Community/document reference

This makes the product credible. Judges will respect that the team knows the difference between AI claims and verified evidence.

5.5. 5. Kashmir Craft Knowledge Graph

Behind every passport, Rootprint stores structured craft knowledge. Over time, this becomes a living graph:

Artisan → Craft → Technique → Material → Region → Product → Buyer Trust

The hackathon can show a simple graph view: multiple passports connected by craft, district, material, and technique. This is the seed of a defensible data moat.

6. MVP: What to Build

6.1. Core MVP

Module	What it must do
Landing page	Strong visual identity: “Give Kashmiri craft a passport.” CTA: Create Passport.
Voice intake	Record audio or upload a prepared sample; select language/craft category.
AI extraction	Convert transcript into structured JSON: artisan, craft, material, technique, story, evidence.
Review screen	Show transcript beside extracted fields; allow edits before publishing.
Scoring engine	Compute a transparent Passport Score out of 100 based on evidence completeness.
Passport generator	Create Passport ID, QR code, shareable URL, and public page.
Public QR page	Mobile-first, premium provenance page with story, evidence, score, and buyer AI.
Admin dashboard	List all passports, view score/status, open QR, copy link, seed demo records.

6.2. MVP Plus: If Time Allows

- **Ask the Craft chat:** question-answering over the passport record.
- **Multilingual summary button:** English, Urdu/Hindi, and one international buyer language.
- **Printable product tag:** Passport ID + QR + “Ask for Rootprint”.
- **Knowledge graph preview:** show how artisans/products connect by craft and technique.
- **Export certificate PDF:** one-click downloadable provenance certificate.

6.3. The Demo Dataset

Prepare 3–5 polished sample passports:

1. **Pashmina shawl** – strongest global recognition and counterfeit problem.
2. **Hand-knotted carpet** – rich technique/process story.
3. **Papier-mache box** – visually appealing and easy for judges to understand.
4. **Walnut wood carving** – material provenance and craft cluster angle.
5. **Sozni embroidery** – precision, time, and maker skill narrative.

7. Technical Architecture

Layer	Implementation idea
Frontend	Next.js/React or equivalent: landing, intake, review, dashboard, public passport page.
Backend API	Create passport, update passport, list passports, generate QR, serve public passport.
Database	Passport records, media links, transcript, extracted JSON, score, QR slug.
AI pipeline	Speech-to-text + LLM structured extraction + summary generation + buyer Q&A over record.
Scoring	Deterministic rule-based score so judges can inspect and trust it.
QR	Unique URL per passport, generated as image/SVG for product tag and page.
Deployment	Public demo URL with seed records and backup transcript path.

7.1. Heritage Record Schema

```
{
  "passport_id": "RP-KMR-PAS-0001",
  "artisan": {
    "name": "",
    "location": "",
    "languages": [],
    "lineage": ""
  },
  "craft": {
    "category": "",
    "techniques": [],
    "tools": [],
    "materials": []
  },
  "product": {
    "name": "",
    "description": "",
    "estimated_time_to_make": "",
    "price_context": ""
  },
  "evidence": {
    "audio_excerpt": "",
    "photos": [],
    "documents": [],
    "review_notes": ""
  },
  "trust": {
    "score": 0,
    "level": "Recorded | Reviewed | Community-backed",
    "last_updated": ""
  }
}
```

7.2. Scoring Model

Signal	Points
Artisan identity and location	15
Craft category and named techniques	20
Material source or provenance detail	15
Lineage, apprenticeship, or process story	15
Photo/audio/video evidence attached	15
Human review completed	10
Community/document reference attached	10
Total	100

8. Pitch Narrative

8.1. Opening

“Everyone in the world has heard the phrase Kashmiri craft. But when a buyer is holding a product, they usually cannot prove three basic things: who made it, where it came from, and whether the story attached to it is real. Rootprint fixes that.”

8.2. The Setup

Hold up or show two product tags:

- Tag A: “Authentic Kashmiri Handmade” – no proof.
- Tag B: Rootprint QR – full passport.

Then say:

“The market is full of Tag A. We are building the world where buyers demand Tag B.”

8.3. The Live Demo

1. Open Rootprint.
2. Create a new passport.
3. Play or record an artisan interview.
4. Show AI extraction into structured fields.
5. Publish the passport.
6. Scan QR with a phone.
7. Ask the passport: “Why should I trust this product?”

8.4. The Close

“Rootprint makes Kashmiri craft impossible to erase and harder to fake. We are not just preserving stories. We are turning them into infrastructure that buyers, artisans, retailers, tourism boards, and institutions can trust.”

9. Business Model

Rootprint begins with the passport. Every future revenue stream attaches to it.

Revenue line	How it works
Freemium passports	Basic passport free; paid upgrades for more products, media, analytics, and printable tags.
Verified review fee	Artisans/businesses pay for expert or community-backed review once partners are onboarded.
QR tag packs	Physical durable tags for shawls, carpets, boxes, workshops, store-fronts, and export packaging.
Export certificate PDFs	Downloadable buyer/export-ready provenance documents.
B2B integrations	Retailers and tourism platforms pay to embed verified Rootprint records.
Data and preservation reports	Aggregated, consent-based craft maps and knowledge preservation insights for institutions.

Why the business can scale

Rootprint does not need to own inventory. It owns the proof layer. Wherever authentic craft is sold, displayed, taught, exported, or researched, the passport has value.

10. Go-to-Market

10.1. Beachhead

Start with one category: **pashmina**. It is globally recognised, frequently counterfeited, emotionally premium, and easy for the jury to understand.

10.2. First Users

- 10–25 artisans, craft businesses, or student-documented maker profiles.
- Local craft sellers willing to attach QR tags.
- University/craft department collaborators who can help review records.
- Tourism-facing locations where scanning a craft story creates instant value.

10.3. Expansion

After pashmina, expand into carpets, papier-mache, walnut wood, embroidery, copperware, and regional craft clusters. The same passport model works for every category; only the extraction prompts, scoring fields, and educational content change.

11. Team Execution: 5 People as 2 + 2 + 1

Unit	Ownership	Deliverables
Team A: 2	Frontend and product experience	Landing page, intake UI, review screen, dashboard, public passport page, mobile polish, QR scan flow.
Team B: 2	Backend and infrastructure	Database schema, API routes, QR generation, scoring service, public passport URLs, deployment, seed data.
Solo: 1	AI, story, demo, and product logic	Prompting, transcript-to-JSON extraction, buyer Q&A, sample craft records, pitch script, demo choreography, final integration quality.

11.1. Integration Milestones

- Milestone 1:** Create dummy passport from frontend and store it in backend.
- Milestone 2:** AI pipeline turns transcript into valid passport JSON.
- Milestone 3:** QR opens public page for a real seeded passport.
- Milestone 4:** Buyer AI answers questions from the passport record.
- Milestone 5:** Full rehearsed demo with backup data and phone scan.

Never depend only on live audio. Keep a polished prerecorded transcript and sample audio ready. The winning demo is the passport transformation, not the microphone.

12. Exact MVP Feature Checklist

- Premium landing page
- Create Passport button
- Audio upload/record UI
- Craft category selector
- Transcript display
- AI extracted JSON
- Editable record fields
- Score out of 100
- Evidence level badge
- Passport ID
- QR code generation
- Public mobile page
- Ask-the-Craft Q&A
- Printable tag view
- Seed records
- Demo mode fallback

13. The Final Pitch in 90 Seconds

Kashmir has one of the strongest craft identities in the world, but the proof of that craft is still fragile. A real artisan may spend years mastering a technique, but at the point of sale, their product often competes with a fake using the same words: authentic, handmade, Kashmiri. Rootprint is the trust layer for Kashmiri craft. An artisan speaks their story in their own

language. Our AI turns that oral knowledge into a structured Craft Passport: who made it, where it came from, what technique was used, what material was used, and what evidence supports the claim. The system generates a QR code that travels with the product. A buyer scans it and can not only read the story, but ask the product questions.

We are starting with pashmina because the counterfeit problem is obvious and the global value is high. But the passport model expands to carpets, papier-mache, walnut wood, embroidery, copperware, tourism, export, and retail integrations.

Rootprint is not a marketplace. It is the proof layer underneath the market. It makes Kashmiri craft impossible to erase and harder to fake.

14. Why This Should Win

Rootprint is rare because it is emotionally powerful, locally grounded, technically credible, commercially scalable, and demoable in one scan.

- **The problem is real:** counterfeit craft, lost memory, weak buyer trust.
- **The solution is tangible:** a QR passport that anyone can scan.
- **The AI is meaningful:** it converts oral knowledge into structured infrastructure.
- **The demo is memorable:** create a passport live, then interrogate it.
- **The Kashmir angle is unbeatable:** the product belongs here first.
- **The moat is obvious:** the earliest structured craft records become the canonical corpus.
- **The business is clean:** passports, tags, verification, certificates, APIs.

Final sentence

Rootprint is how Kashmir stops being a label on a product and becomes a verifiable identity owned by its makers.

ROOTPRINT – The Trust Layer for Kashmiri Craft

Built so the maker's name travels with the work.